

interval is set sufficiently large such that the Administrator can synchronize all Nodes in the operational group with the new epoch key list within that time.

The rotation rate for epoch keys is expected to be on the order of days to weeks for typical applications, but the rate is configurable as required by the key distribution Administrator. Because of the relatively long rotation interval, and the overlap of active epoch keys, local clock drift within Nodes is generally not a concern.

#### 4.17.3.4. Epoch Key Rotation without Time Synchronization

Although epoch keys are distributed with an associated start time, it is nonetheless possible for Nodes that do not maintain a synchronized clock to participate in key rotation. Specifically, upon receiving a new epoch key list from the key distribution Administrator, such a Node can note which of the keys is the current epoch key by comparing their relative start times and using the current epoch key which has the second newest time. It can then use the current key for all locally initiated security interactions until such time as it makes contact with the distribution Administrator again.

This scheme requires the Node to receive epoch keys from the key distribution Administrator at a rate that is at least as fast as the configured *key propagation interval*. The Administrator SHOULD provide a sufficient set of epoch keys to Nodes that do not maintain synchronized time so that they can maintain communication with other group members while a key update is in progress. The key distribution Administrator SHOULD update all Nodes without time, such as ICDs, before the *new epoch key* is activated, and then let Nodes with time all roll to the *new epoch key* at the synchronized start time.

#### 4.17.3.5. Epoch Key Rotation logic

The full life-cycle of Epoch Key rotation is shown in [Figure 18, “Epoch Key Rotation”](#). For each epoch key slot, the start time of the key is shown as one of the following values:

- **New** - a key with a start time in the future.
- **Current** - the active key with the newest start time.
- **Previous** - the active key with the second newest start time.
- **Old** - an active key with the third newest start time.

The diagram shows two types of state transitions:

1. **Admin** - an update of an old key by the Administrator. Changes made during this state transition are indicated in green.
2. **Epoch Activate** - activation of an epoch key due to system time becoming greater than the start time. Changes during this state transition are indicated in yellow.